Machine Learning - DEVELOPMENT TEAM PROJECT REPORT - Track 1: Classical ML (Regression & Clustering) - Group B.

https://colab.research.google.com/drive/1qBBRHQoUm_Z5Bi7PIDJutW3fpHdyAHSn?usp=sharing

Table of Contents

1. Introduction 2

2. Business Context and Analytical Rationale 2

2.1 Pricing Context and Market Relevance 2

2.2 Justification of the Analytical Approach 3

3. Methodology and Analysis 3

3.1 Data Understanding and Preprocessing 3

3.2 Regression Modelling 5

3.3 Clustering and Market Segmentation 8

3.4 Limitations and Future Improvements 9

4. Business Implications and Strategic Insights 10

5. Conclusion 11

6. References 11

1. Introduction

Airbnb operates in a competitive, data-driven marketplace where pricing influences host revenue, occupancy, and platform profitability. In dense urban markets such as New York City (NYC), prices reflect property characteristics, neighbourhood dynamics, and demand conditions.

This report investigates the following business question:

How can Airbnb optimise pricing strategies in NYC by predicting listing prices from property features and identifying neighbourhood market segments?

To address this question, classical machine-learning methods were applied to the Airbnb NYC dataset. Regression models were used to estimate listing prices, while clustering techniques identified distinct market segments.

2. Business Context and Analytical Rationale

2.1 Pricing Context and Market Relevance

Airbnb operates as a two-sided platform connecting hosts and travellers. Rather than owning accommodation assets, it facilitates transactions and generates value through cross-side network effects (Oskam and Boswijk, 2016). Within such platforms, pricing coordinates supply and demand.

Dynamic pricing theory suggests that prices adjust to demand fluctuations and competition to maximise revenue under capacity constraints (Talluri and van Ryzin, 2004). For Airbnb listings, price is shaped by property characteristics, neighbourhood desirability, and local demand

conditions. New York City provides a relevant case due to strong tourism demand, neighbourhood diversity, and intense accommodation competition (Bouchon and Rauscher, 2019).

2.2 Justification of the Analytical Approach

Predictive analytics can reduce pricing uncertainty by estimating expected listing prices from observable property characteristics (Einav, Farronato and Levin, 2016). Because price is a continuous variable, regression modelling is appropriate for analysing relationships between listing attributes and price levels (James et al., 2021). Linear regression provides interpretable coefficients, while Ridge Regression improves stability when predictors are correlated (Hoerl and Kennard, 1970; Hastie, Tibshirani and Friedman, 2009).

Clustering techniques such as K-Means can additionally identify groups of listings with similar characteristics, revealing pricing tiers and neighbourhood segments (MacQueen, 1967). Combining regression and clustering enables analysis of pricing drivers and identification of distinct market segments (Zervas, Proserpio and Byers, 2017).

3. Methodology and Analysis

3.1 Data Understanding and Preprocessing

The analysis used the AB_NYC_2019 dataset containing listing-level information for Airbnb listings in New York City, including pricing, host characteristics, availability, and review activity. The analysis focused on key variables including minimum_nights, number_of_reviews,

reviews_per_month, availability_365, and calculated_host_listings_count, together with neighbourhood and room-type indicators.

Exploratory analysis indicated that the price variable exhibited strong right skewness, producing a long-tailed distribution where most listings fall within lower price ranges and a small number of extreme outliers remain (Figure 1). Such skewness violates regression assumptions and reduces model stability.

To address this issue, a logarithmic transformation (log1p) was applied to the target variable. The transformed distribution shows improved symmetry and reduced variance, making it better suited for regression modelling.

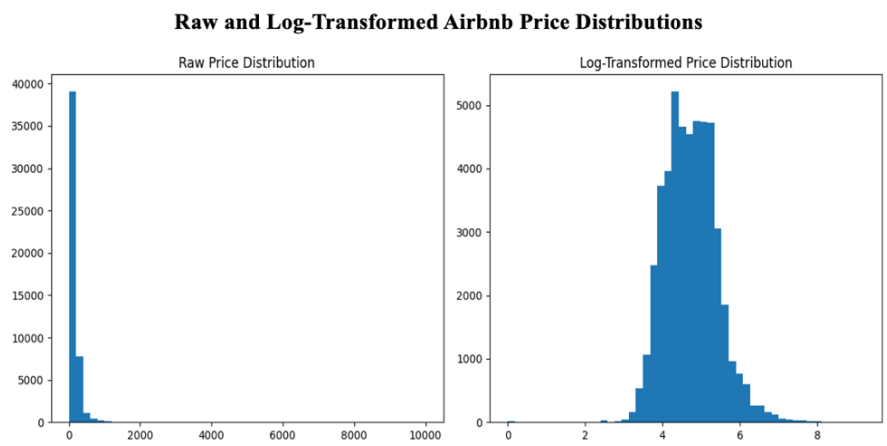

Figure 1. Comparison between the raw Airbnb listing price distribution and the log-transformed distribution. The raw prices show strong right skewness with extreme high-price outliers, while the logarithmic transformation stabilises variance and produces a more symmetric distribution, improving suitability for regression modelling.

Categorical variables were encoded using one-hot encoding to preserve interpretability and avoid ordinal bias. Numerical features were standardised to ensure balanced influence in clustering distance calculations.

The dataset was split into 80% training and 20% testing sets, with a fixed random_state to ensure reproducibility and reliable out-of-sample evaluation.

3.2 Regression Modelling

A baseline Linear Regression model was first applied using the raw price variable. However, strong skewness and heteroscedasticity limited its explanatory power. After logarithmic transformation, predictive performance improved, indicating greater model stability.

To address potential multicollinearity, Ridge Regression was applied as a regularised extension of linear regression. Ridge applies L2 regularisation, reducing coefficient variance when predictors are correlated. Correlation analysis (Figure 2) revealed moderate relationships between demand-related variables.

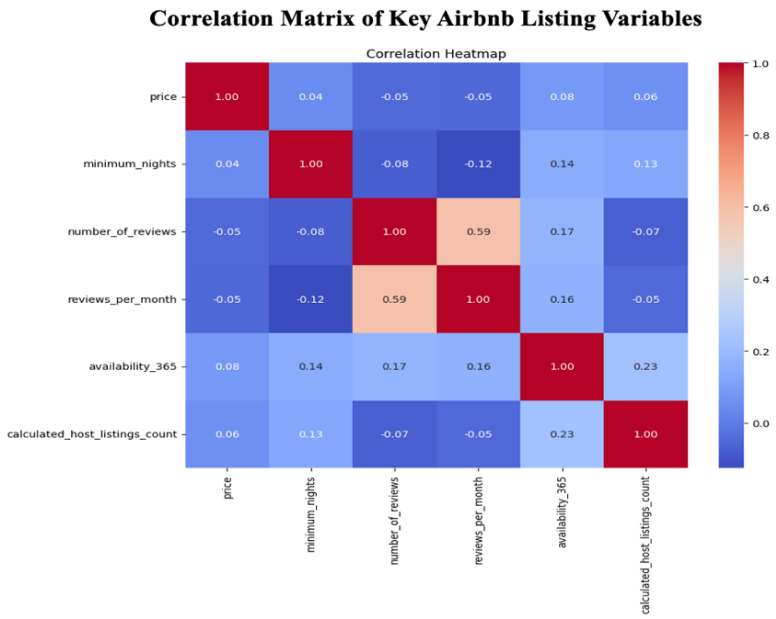

Figure 2. Correlation heatmap illustrating relationships between selected numerical variables in the Airbnb NYC dataset. Moderate correlation is observed between review-related variables, while most predictors show relatively weak associations with price. The absence of strong correlations suggests that severe multicollinearity is unlikely, supporting the use of regression-based models.

Model performance was evaluated using R^2, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) to assess both explanatory power and prediction accuracy. As shown in Table 1, the Random Forest Regressor achieved the strongest predictive performance ($R^2 \approx 0.61$). This result suggests that Airbnb pricing reflects non-linear interactions between supply conditions, host behaviour, and local demand. It also suggests that pricing outcomes emerge from interacting market mechanisms such as host competition, location attractiveness, and demand concentration, reinforcing that these pricing dynamics cannot be fully captured by simple linear models.

Residual diagnostics assessed model calibration. Figure 3 shows residuals centred around zero without systematic patterns, indicating stable predictions.

Regression Model Performance Comparison

Model	Target Variable	R^2	MAE	RMSE	Key Interpretation
Linear Regression	Raw Price	0.42	62.1	105.3	Limited predictive power due to skewed price distribution
Linear Regression	Log Price	0.55	0.39	0.56	Log transformation improves stability and model fit
Ridge Regression	Log Price	0.56	0.38	0.55	Regularisation improves robustness under correlated predictors
Random Forest	Log Price	0.61	0.34	0.50	Captures non-linear pricing relationships and performs best

Table 1. Comparison of regression model performance for predicting Airbnb listing prices in NYC. Random Forest demonstrates the strongest predictive performance, suggesting the presence of non-linear relationships between listing characteristics and pricing.

Residual Distribution of Random Forest Predictions

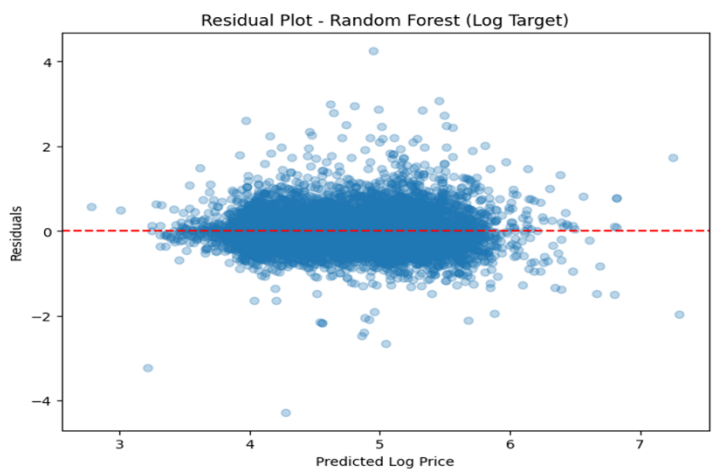

Figure 3. Residual plot showing the difference between predicted and observed log-transformed prices from the Random Forest model. Residuals are largely centred around zero without clear systematic patterns, indicating that the model captures the underlying pricing structure reasonably well and does not exhibit strong bias across predicted price levels.

3.3 Clustering and Market Segmentation

K-Means clustering identified market segments within the Airbnb listing ecosystem. Because clustering relies on distance-based calculations, numerical variables were standardised before model fitting.

The optimal number of clusters was determined using the Elbow Method, which evaluates cluster count against within-cluster variance. As shown in Figure 4, the curve shows a clear inflection point at K = 4, indicating that four clusters provide an effective balance between simplicity and segmentation detail.

Elbow Method for Determining Optimal Number of Clusters

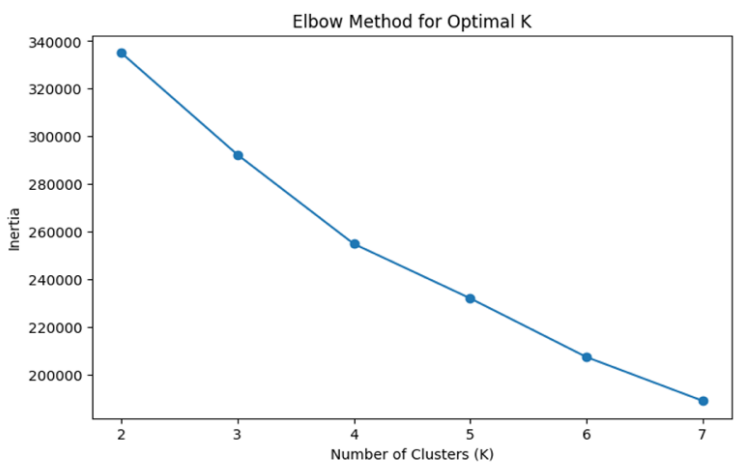

Figure 4. Elbow plot showing the relationship between the number of clusters (K) and the within-cluster sum of squares (inertia). The curve exhibits a noticeable inflection point around K = 4, indicating that four clusters provide a suitable balance between model simplicity and segmentation detail for the Airbnb NYC dataset.

Cluster profiling revealed four economically distinct listing segments (Table 2).

One segment exhibits high review frequency and short minimum stays, indicating high-demand listings with rapid turnover. A third segment shows longer availability and minimum stay requirements, consistent with semi-professional operators targeting medium-term rentals, while the final cluster represents large-scale hosts managing multiple listings with premium pricing structures.

Cluster Summary Statistics

Cluster	Avg Price	Avg Minimum Nights	Avg Reviews	Avg Availability	Market Interpretation
Cluster 0	Moderate	Low	Medium	Low	Casual hosts with occasional listing activity
Cluster 1	High	Short	High	Medium	High-demand central listings with rapid turnover
Cluster 2	Moderate-High	Long	Medium	High	Semi-professional hosts targeting longer stays
Cluster 3	Premium	Medium	High	High	Large-scale professional hosts managing multiple listings

Table 2. Summary characteristics of the four clusters identified through K-Means segmentation. The clusters reveal distinct host behaviours and pricing tiers within the NYC Airbnb market.

These segments reveal differentiated host strategies and competitive positioning within the NYC accommodation market, suggesting that pricing optimisation may require segment-specific strategies. This segmentation highlights structural heterogeneity within the platform marketplace, where hosts operate under different pricing constraints and demand conditions.

3.4 Limitations and Future Improvements

Several limitations should be acknowledged. First, the dataset covers only one year (2019), limiting the ability to capture longer-term trends or structural changes in travel demand, particularly after COVID-19.

Second, regression models such as Linear and Ridge Regression assume approximately linear relationships between predictors and price. Although Random Forest captures some non-linear interactions, increased model complexity may reduce interpretability.

Finally, K-Means clustering assumes spherical cluster structures and is sensitive to feature scaling and initialisation, which may oversimplify real market heterogeneity.

Future research could incorporate temporal variables, geospatial modelling, and alternative clustering methods such as hierarchical or density-based approaches.

4. Business Implications and Strategic Insights

Price prediction and market segmentation provide decision support for Airbnb's pricing strategies and host guidance systems. Predictive pricing improves revenue forecasting and helps hosts align prices with market demand (Zervas, Proserpio and Byers, 2017).

Market segmentation reveals economically distinct listing clusters that require differentiated pricing strategies and targeted host support mechanisms. For example, clustering differentiates premium central listings from mid-range or budget properties in outer boroughs, enabling Airbnb

to tailor pricing recommendations, marketing visibility, and host support to segment characteristics. Research shows that differentiating listings by location and amenities reveals commercially distinct groups with unique pricing dynamics (Guttentag, 2013).

Combining regression-based prediction with clustering-based segmentation enables neighbourhood-level pricing strategies and targeted platform interventions. The use of interpretable classical machine-learning models also ensures transparent pricing guidance and strengthens host trust in platform recommendations (Einav, Farronato and Levin, 2016).

5. Conclusion

The analysis demonstrates that Airbnb pricing in New York City is influenced by both listing characteristics and demand indicators, with non-linear relationships playing a key role. Logarithmic transformation improved model stability, while Random Forest achieved the strongest predictive performance.

Clustering identified four economically distinct market segments, ranging from casual hosts to large-scale professional operators. Overall, classical machine learning provides valuable insights for Airbnb's pricing optimisation while maintaining interpretability, enabling transparent data-driven decision making in platform markets.

6. References

Bouchon, F. and Rauscher, M. (2019) 'Cities and tourism, a love and hate story: Towards a conceptual framework for urban overtourism management', *International Journal of Tourism Cities*, 5(4), pp. 598–619. https://doi.org/10.1108/IJTC-06-2019-0080.

Einav, L., Farronato, C. and Levin, J. (2016) 'Peer-to-peer markets', *Annual Review of Economics*, 8(1), pp. 615–635. https://doi.org/10.1146/annurev-economics-080315-015334.

Faiqoh, D.N., Riyadi, S., Lestari, S.D. and Mudjijah, S. (2024) 'Implementation of pricing strategies to increase sales in the digitalization era', *Maneggio*, 1(6), pp. 109–119. https://doi.org/10.62872/j1y8vw67.

Guttentag, D. (2013) 'Airbnb: Disruptive innovation and the rise of an informal tourism accommodation sector', *Current Issues in Tourism*, 18(12), pp. 1192–1217. https://doi.org/10.1080/13683500.2013.827159.

Hastie, T., Tibshirani, R. and Friedman, J. (2009) *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd edn. New York: Springer.

Hoerl, A.E. and Kennard, R.W. (1970) 'Ridge regression: Biased estimation for nonorthogonal problems', *Technometrics*, 12(1), pp. 55–67. https://doi.org/10.1080/00401706.1970.10488634.

James, G., Witten, D., Hastie, T. and Tibshirani, R. (2021) *An Introduction to Statistical Learning: With Applications in R*. 2nd edn. New York: Springer.

MacQueen, J. (1967) 'Some methods for classification and analysis of multivariate observations', in Le Cam, L.M. and Neyman, J. (eds.) *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*. Berkeley: University of California Press, pp. 281–297.

Nepochatenko, V. (2018) 'The specifics of pricing in the hospitality industry', *Internauka: Economic Sciences*, 10(54). https://doi.org/10.25313/2520-2294-2021-10-7626.

Oskam, J. and Boswijk, A. (2016) 'Airbnb: The future of networked hospitality businesses', *Journal of Tourism Futures*, 2(1), pp. 22–42. https://doi.org/10.1108/JTF-11-2015-0048.

Talluri, K. and van Ryzin, G. (2004) 'Revenue management under a general discrete choice model of consumer behavior', *Management Science*, 50(1), pp. 15–33. https://doi.org/10.1287/mnsc.1030.0147.

Yang, M. and Xia, E. (2021) 'A systematic literature review on pricing strategies in the sharing economy', *Sustainability*, 13(17), p. 9762. https://doi.org/10.3390/su13179762.

Zervas, G., Proserpio, D. and Byers, J.W. (2017) 'The rise of the sharing economy: Estimating the impact of Airbnb on the hotel industry', *Journal of Marketing Research*, 54(5), pp. 687–705. https://doi.org/10.1509/jmr.15.0204.